

athlete clearing the hurdle too high relative to the hurdle height and with too much knee flexion.) Keep the hips and shoulder square and the knees and feet together without lateral deviation.

Downward movement: Rotate over the hurdle so the lean is in the direction of the subsequent jump on landing. Dorsiflex the ankles in preparation for an active flat-foot landing on the opposite side of the barrier, and immediately repeat the jump to the starting side. (*Note:* The third photo does not show the athlete leaning back toward the starting side for the return jump.)

Note: The intensity level of the lateral barrier hop can be increased from medium to high by progressively increasing the height of the barrier or by performing movement with one leg only (i.e., a hop), which is an excellent exercise for aiding direction change capacity in the athlete. By reversing the direction of movement, the ground contact time and the amortization phase of the movement are changed compared to movements in the same direction. This will be influenced further by the height of the hurdle.

COMMON ERRORS

- **Not maintaining a square hip and shoulder position, so the jump does not stay in the frontal plane**
- **Lack of preparatory lean in both takeoff (center of mass should lean in the direction of travel) and landing (orient the body so that it is leaning back toward the start for the return jump), which increases the ground contact time and reduces the effectiveness of the drill**



19 FOUR-HURDLE HOP

Intensity level: High

Direction: Lateral and vertical

Equipment: Four hurdles: two pairs of hurdles. Each hurdle in the pair is separated by 12 inches (30 cm). Each pair is separated by 18 inches (46 cm).

Starting position: Get into a comfortable, upright stance on the right foot; the line of four hurdles will be to the left. The non-jumping leg is held in a stationary position with the hip and knee flexed during the exercise.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Movements:

1. Lean in the direction of travel.
2. Explosively hop left over the first hurdle with the right foot, using both arms to assist.
3. Dorsiflex the ankles in preparation for an active flat-foot landing.
4. Land on the right foot, and immediately repeat the hop over the next hurdle.
5. Land on the right foot, and immediately repeat the hop over the next two hurdles.
6. Land on the left foot, and immediately hop right over the first hurdle with the left foot, using both arms to assist.
7. Land on the left foot, and immediately repeat the hop over the next three hurdles; finish by landing on the right foot.

COMMON ERRORS

- **Not reflecting subsequent takeoff direction in the body lean (i.e., the center of mass in the takeoff and the landing before direction change should reflect the direction of travel in the subsequent movement; failure to do so increases the ground contact time and reduces the effectiveness of the drill)**
- **Not increasing ground contact time for both the two-hurdle hop and the change of direction movement, meaning insufficient force is produced in the jump**



20

PROGRESSIVE LATERAL JUMP

Intensity level: High

Direction: Diagonally lateral and vertical

Equipment: String tied at athlete's eye level and loosely secured to the ground at a 45-degree angle to the ground

Starting position: Get into a comfortable, upright stance on one side of the string next to the anchor on the ground.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Movements:

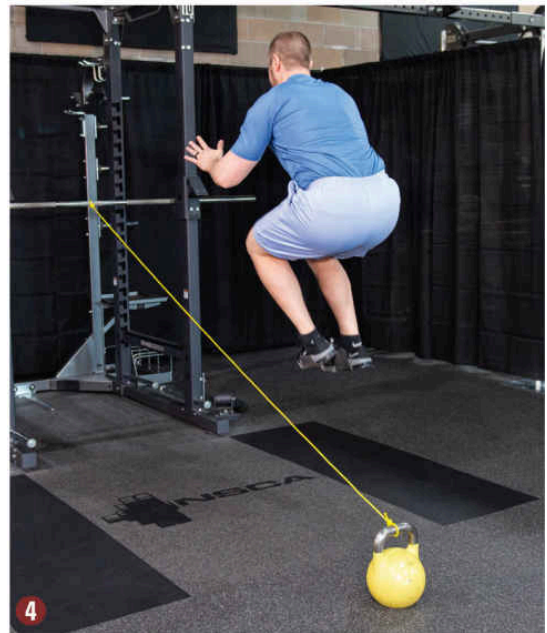
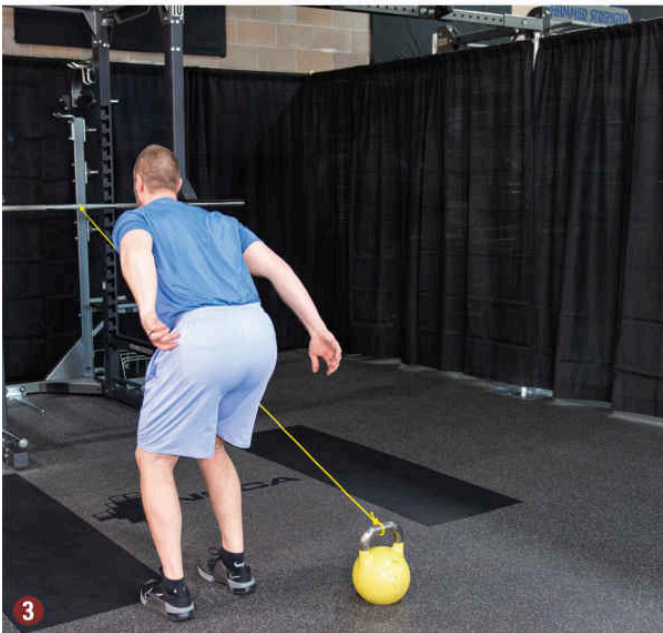
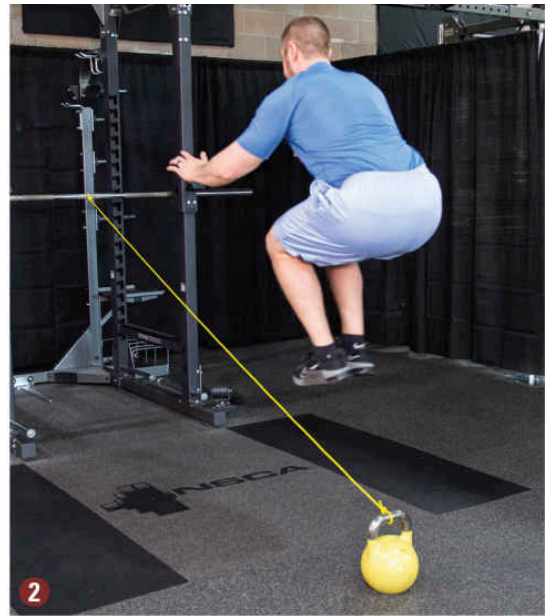
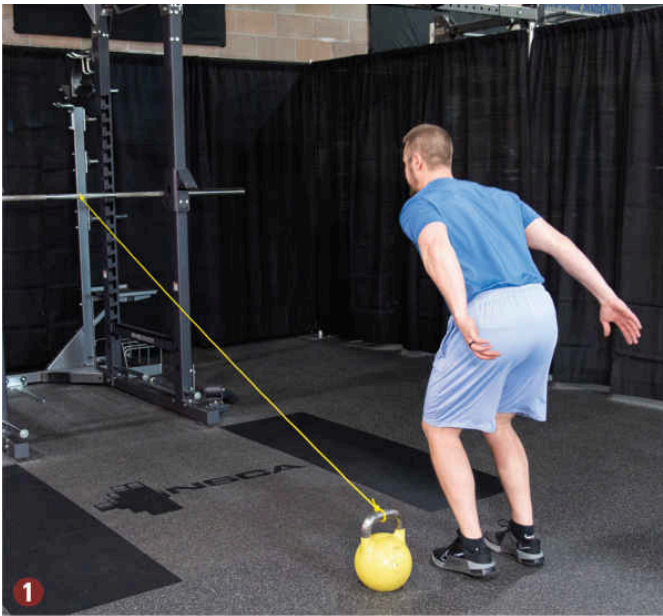
1. Explosively jump diagonally (sideways and forward) over the string using both arms to assist the movement.

2. Dorsiflex the ankles in preparation for an active flat-foot landing.
3. Land on the other side of the string, and immediately repeat the jump diagonally forward. Continue the jumping sequence over the string as it gets progressively higher.

Note: As the string gets higher, the athlete will need a longer ground contact time and more hip and knee flexion to clear the string. Ensure that the anchor on the string enables the string to give way should the athlete make contact with it.

COMMON ERRORS

- **Changing the orientation of the hips (i.e., rotate in the transverse plane) during takeoff, which increases the ground contact time on the subsequent jump**
- **Not increasing ground contact time and hip and knee flexion through the sequence, meaning insufficient force is produced as the required jump height increases**



Bounds

21 SKIP

Intensity level: Low

Direction: Horizontal and vertical

Starting position: One leg is lifted to approximately 90 degrees of hip and knee flexion with the ankle dorsiflexed.

Arm action: Reciprocal (as one leg is lifted, the opposite arm is lifted)

Preparatory movement: Begin with a countermovement on one leg.

Upward movement: Drive up and forward on one leg. The opposite leg should remain in the starting flexed position until landing.

Downward movement: Land in the starting position on the same leg. Immediately repeat the skip with the opposite leg.

COMMON ERRORS

- **Rotating at the hip and shoulders during the movement (which should be in the sagittal plane only), detracting from the velocity achieved**
- **Coordination errors in the reciprocal arm and leg action, which detract from the forward momentum achieved**



22

POWER SKIP

Intensity level: Low

Direction: Vertical and horizontal

Starting position: One leg is lifted to approximately 90 degrees of hip and knee flexion with the ankle dorsiflexed.

Arm action: Double arm

Preparatory movement: Begin with a countermovement on one leg.

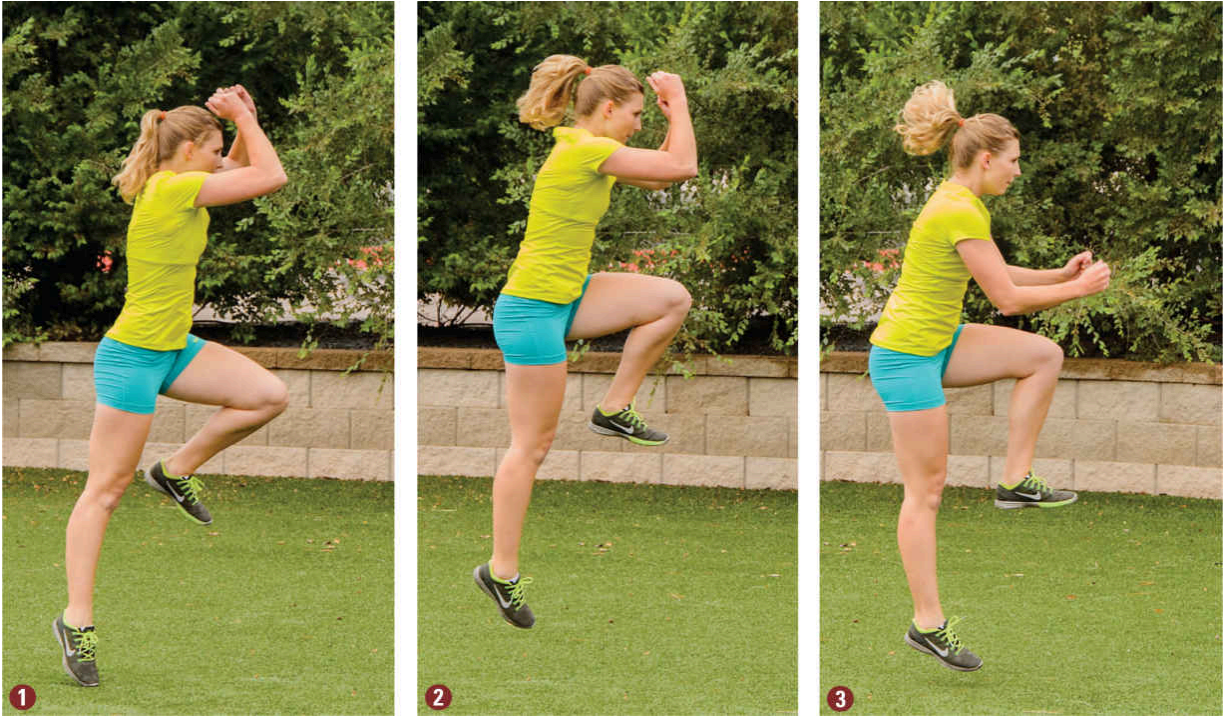
Upward movement: Drive up and forward on one leg. Move the flexed, non-jumping leg up and into greater hip and knee flexion while jumping. Both arms should be used to assist with the upward movement.

Downward movement: Land in the starting position on the same leg. Immediately repeat the skip with the opposite leg.

Note: Emphasis is on high-quality ground contact and leg drive as athletes determine the effectiveness of the skip. This movement should be more vertical and with a faster ground contact than the higher intensity linear bounding drills listed later in the chapter.

COMMON ERRORS

- **Not attacking the ground with significant intensity to accelerate the center of mass forward in subsequent takeoff; this influences the subsequent flight phases and landings**
- **Coordination errors in the double-arm action, which detracts from the forward momentum achieved**



23 LATERAL SKIP

Intensity level: Medium

Direction: Vertical and lateral

Starting position: One leg is lifted to approximately 90 degrees of hip and knee flexion with the ankle dorsiflexed. This leg should be opposite to the direction of movement (i.e., if the skip is to the right, the left leg should be raised).

Arm action: Reciprocal (as one leg is lifted, the opposite arm is lifted)

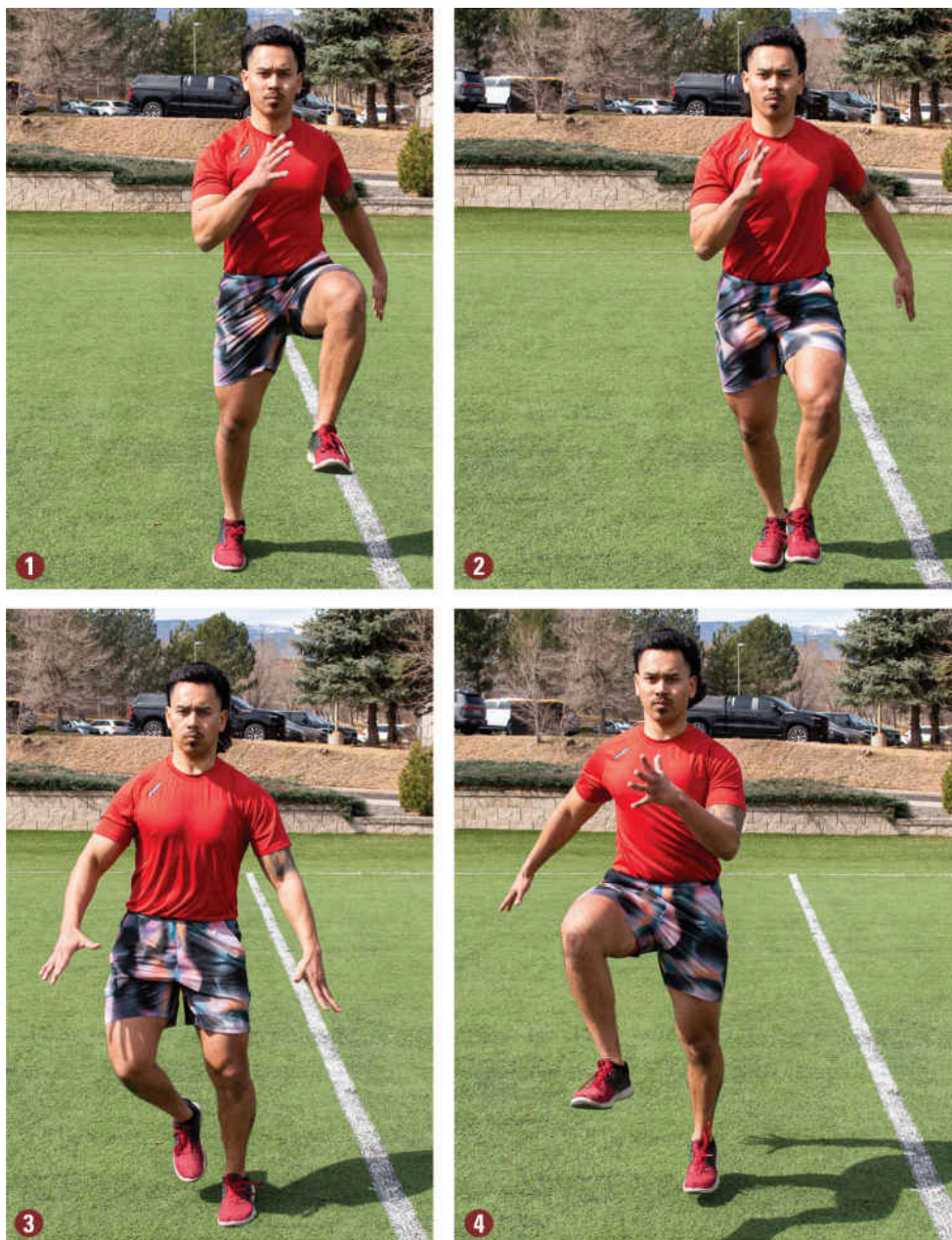
Preparatory movement: Drive the lifted leg forcefully toward the ground and away from the direction of movement so that it contacts the ground outside of the base of support.

Upward movement: The lateral drive of the downward leg will cause the body to move up and sideways away from the foot contact. As this happens, the other leg should flex at the hip and knee in flight. The drive leg typically takes a small adjustment step to aid stability and rhythm in the action.

Downward movement: Land in the starting position on the same leg. Immediately repeat the skip, continuing the lateral movement for a number of repetitions or over a given distance, then use the opposite leg.

COMMON ERRORS

- **Not contacting the ground with sufficient lateral distance from the hip to cause sideways momentum to occur**
- **Not controlling rotations at the hip and shoulders during the movement, which detracts from the lateral velocity achieved**



24 LINEAR BOUND

Intensity level: Medium

Direction: Horizontal and vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Single arm

Preparatory movement: Jog at a comfortable pace; begin the drill with the left foot forward.

Upward movement: Push off with the left foot as it contacts the ground. During push-off, bring the right leg forward by flexing the thigh to a position approximately parallel with the ground and the knee at 90 degrees, with the ankle stiff and dorsiflexed.

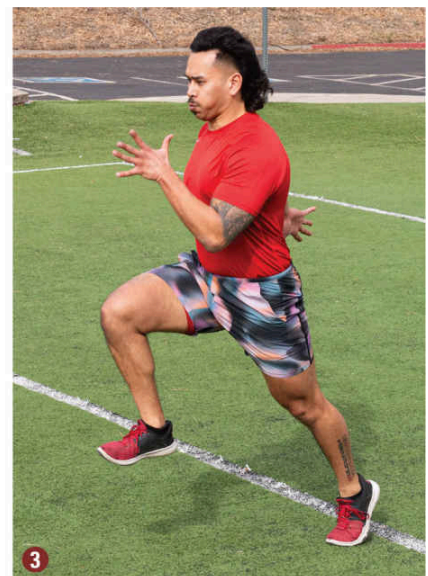
During this flight phase of the drill, reach forward with the left arm.

Downward movement: Attack the ground with an active flat-foot contact. Land on the right leg, and immediately repeat the sequence on the opposite side.

Note: A linear bound is an exaggeration of the running gait; the goal is to cover as great a distance as possible during each stride.

COMMON ERRORS

- **Not achieving sufficient horizontal velocity to initiate the driving action of the bound**
- **Not achieving sufficient height with the lead leg in the flight phase to attack the ground with enough force to accelerate the athlete forward**
- **Overreaching with the lead leg and then not driving down under the hips**



25

DOUBLE-ARM LINEAR BOUND

Intensity level: Medium

Direction: Horizontal and vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Jog at a comfortable pace; begin the drill with the right foot forward.

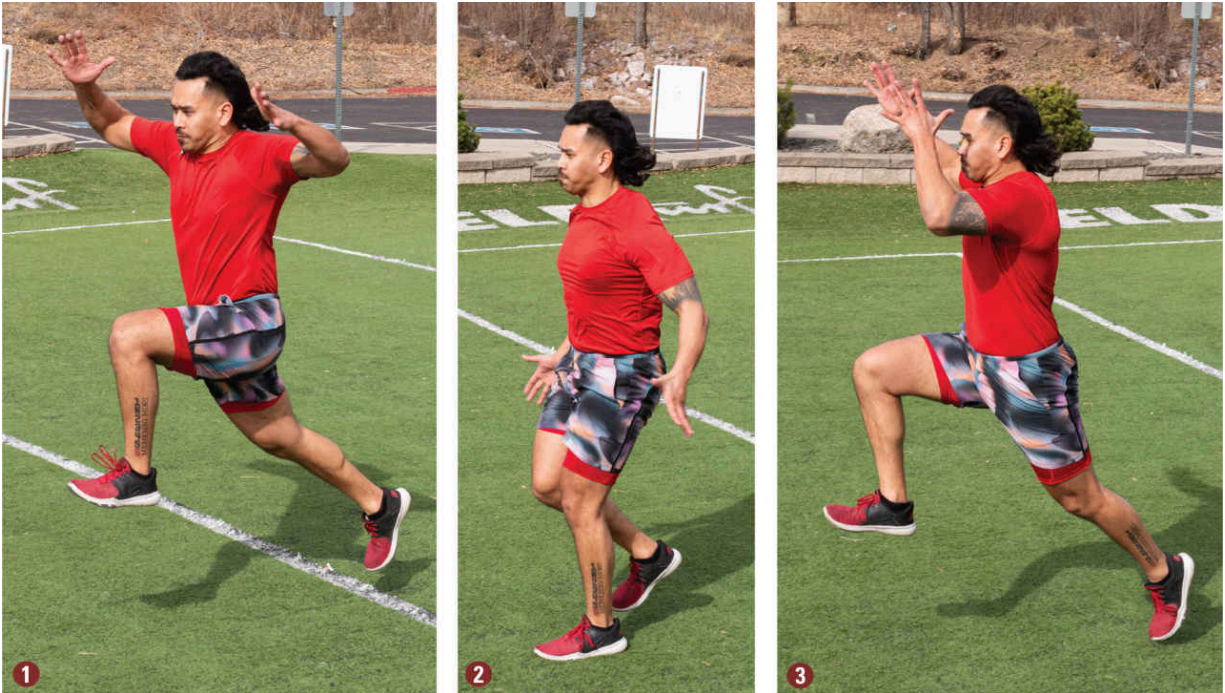
Upward movement: Push off with the right foot as it contacts the ground. During push-off, bring the left leg forward by flexing the thigh to a position approximately parallel with the ground, the knee at 90 degrees, and the ankle dorsiflexed. During this flight phase of the drill, reach forward with both arms. Sequence the arm swing so that they come forward from behind the body into landing, and forward from the hips to reach in front of the body during takeoff.

Downward movement: Land with an active flat-foot on the left leg and immediately repeat the sequence on the opposite side.

Note: A bound is an exaggeration of the running gait; the goal is to cover as great a distance as possible during each stride.

COMMON ERRORS

- **Not achieving sufficient horizontal velocity to initiate the driving action of the bound**
- **Coordination errors between arm and leg action; both arms should be forward as the athlete takes off and behind as the athlete lands**
- **Overreaching with the lead leg**



Box Drills

26 DROP STEP TO LATERAL BOUND AND RETURN

Intensity level: High

Direction: Lateral and vertical

Equipment: Plyometric box, 6 to 18 inches (15-46 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: None

Preparatory movement: Step from the box with the right foot, ankles dorsiflexed.

Downward movement: Land on the ground with active flat foot contact in the right foot.

Upward movement: Immediately push up and sideways toward the left. Land on the left foot and immediately push back toward the

right. Land on the right foot in front of the box and immediately stabilize the landing, with hips and knees aligned with toes.

Note: The trunk orientation should change in the flight so that the athlete is leaning toward the intended direction of travel into the landing.

COMMON ERRORS

- **Not orienting the body in landing to facilitate optimal lateral takeoff**
- **Absorbing the landing force through excessive hip and knee flexion, prolonging the ground contact time**
- **Crossing the foot of the non-landing leg behind the landing leg**



27

SINGLE-LEG PUSH-OFF

Intensity level: Low

Direction: Vertical

Equipment: Plyometric box, 6 to 18 inches (15-46 cm) high

Starting position: Stand facing the plyometric box with one foot on the ground and one foot on the box. The heel of the foot on the box should be near the box's closest edge.

Arm action: Double arm

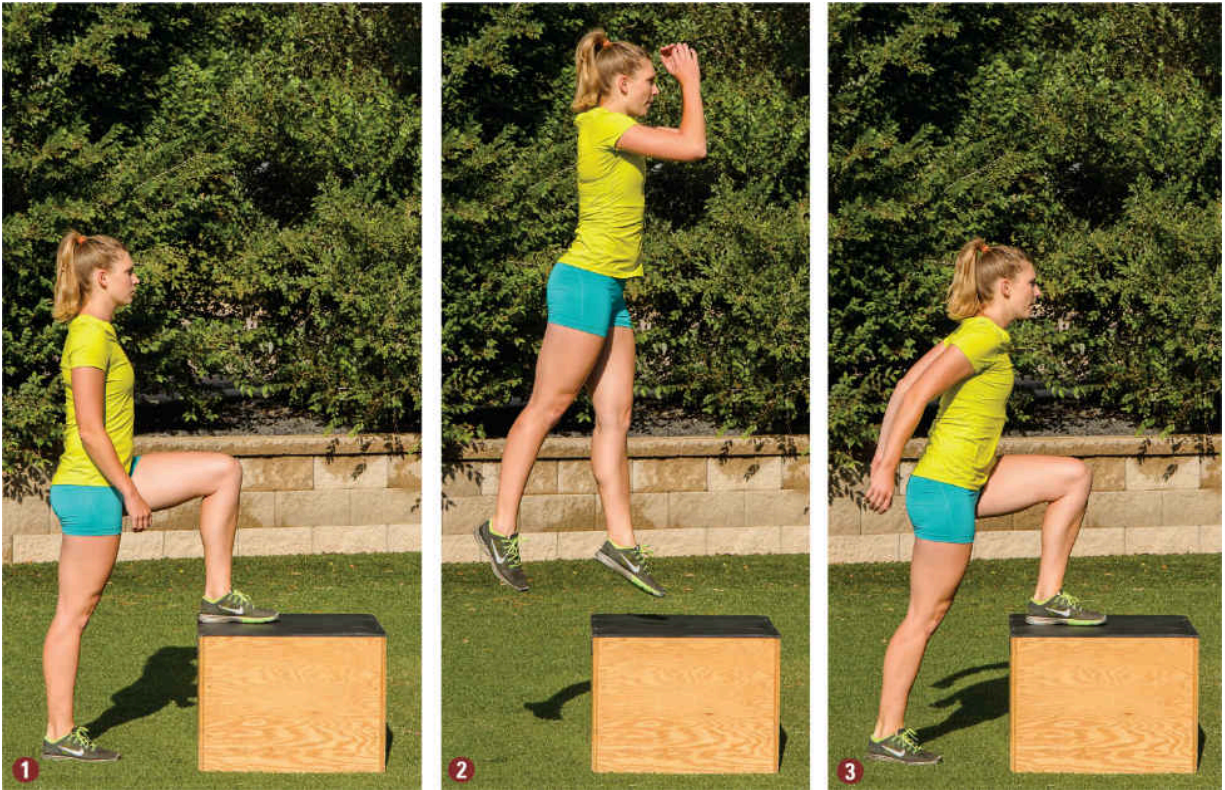
Preparatory movement: Lift the front foot slightly so that it can be driven down into the box.

Upward movement: Drive the front foot down into the box to cause an upward jumping action.

Downward movement: Land with the same foot on the box; this foot should land just before the ground foot. Immediately repeat the movement.

COMMON ERRORS

- **Not orienting the body at takeoff to optimally generate vertical force**
- **Not creating sufficient vertical force with the front foot**



28

ALTERNATING-LEG PUSH-OFF

Intensity level: Low

Direction: Vertical

Equipment: Plyometric box, 6 to 18 inches (15-46 cm) high

Starting position: Stand facing the plyometric box with one foot on the ground and one foot on the box. The heel of the foot on the box should be near the box's closest edge.

Arm action: Double arm

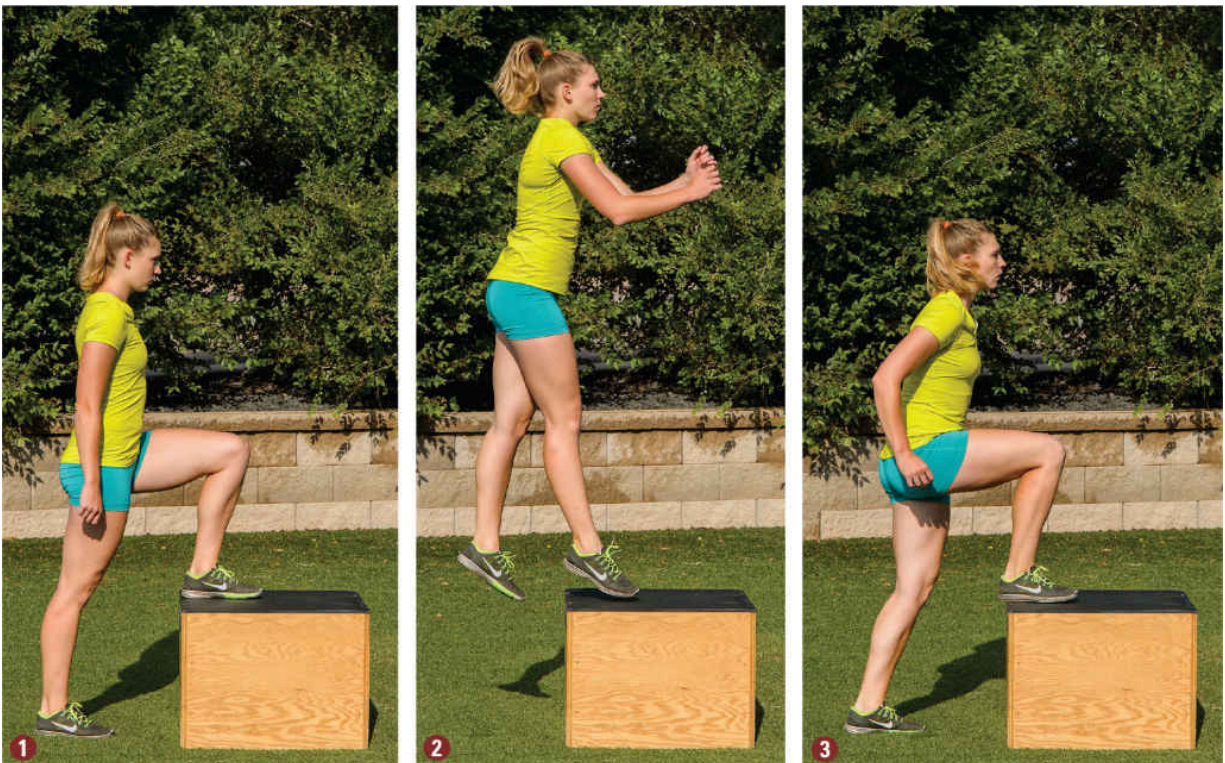
Preparatory movement: Lift the front foot slightly so that it can be driven down into the box.

Upward movement: Drive the front foot down into the box to cause an upward jumping action.

Downward movement: Land with the opposite foot on the box; this foot should land just before the ground foot. Immediately repeat the movement, reversing the feet each repetition.

COMMON ERRORS

- **Not creating sufficient vertical force to enable the switch of legs to occur above the box**
- **Asynchronous actions between the front leg and back leg, which means that the high knee position at the end of the jump does not occur at the point of maximum height or leg extension; this results in poor landing positions that prolong ground contact times for the next jump**



29

LATERAL PUSH-OFF

Intensity level: Low

Direction: Vertical

Equipment: Plyometric box, 10 to 18 inches (25-46 cm) high

Starting position: Stand to one side of the plyometric box with one foot on the ground and one foot on the box. The inside of the foot on the box should be near the box's closest edge.

Arm action: Double arm

Preparatory movement: None or slightly lift the foot off the box.

Upward movement: Jump up using the foot on the box to push off.

Downward movement: Land with the same foot on the box; this foot should land just before the ground foot. Immediately repeat the movement.

COMMON ERRORS

- **Not producing sufficient vertical force from the front foot**
- **Poor coordination or strength deficiencies, which means that peak height of the center of mass does not occur at the same time as maximal extension of the lower limbs and flexion in the shoulders**



30 REAR FOOT–ELEVATED MULTIDIRECTIONAL HOP

Intensity level: Medium

Direction: Vertical and lateral

Equipment: Plyometric box, 6 to 18 inches (15-46 cm) high; mini-hurdle, 6 to 24 inches (15-61 cm) high

Starting position: Start in a deep split squat position, with the rear foot elevated on a plyometric box (or equivalent). The foot on the ground should rest parallel to the mini-hurdle.

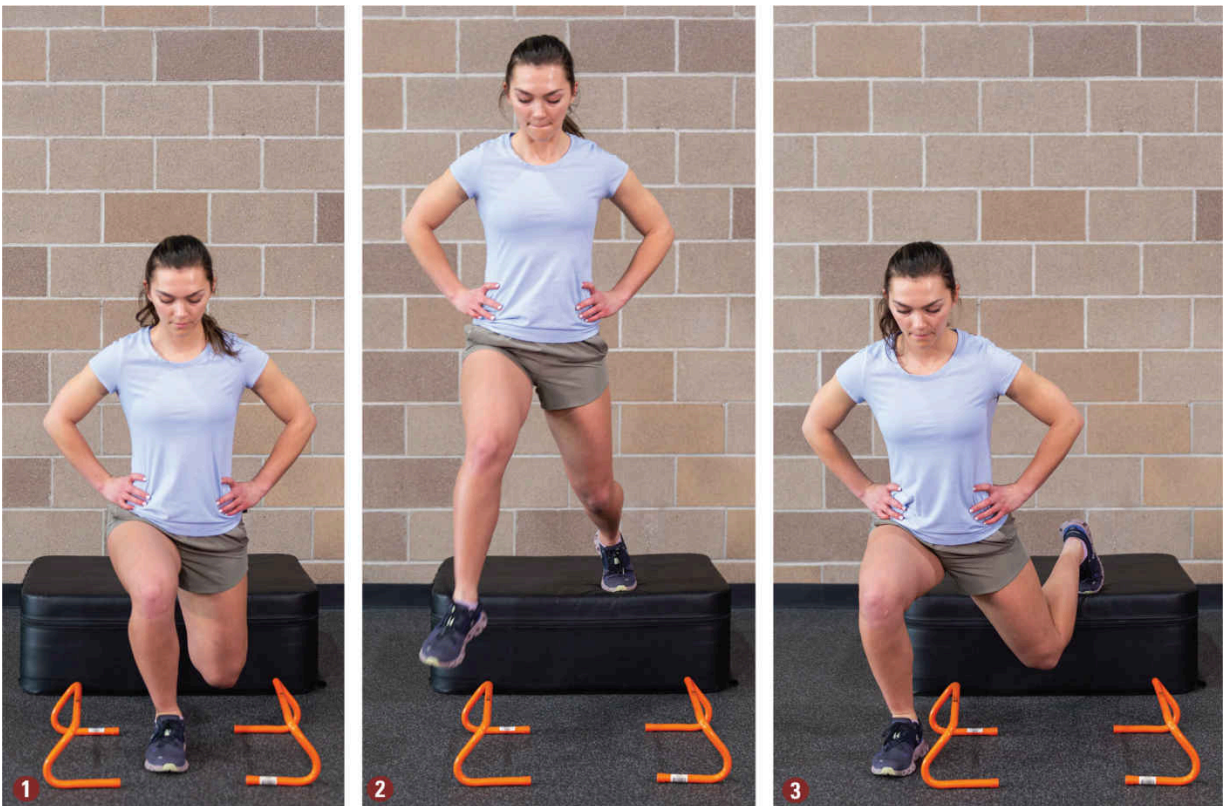
Preparatory movement: None

Upward movement: Push off with the standing leg to rise vertically and sideways over the mini-hurdle. Keep the ankle dorsiflexed in preparation for an active flat-foot contact with the ground on landing.

Downward movement: Land with the same foot on the opposite side of the hurdle. Immediately repeat the movement to the opposite side.

COMMON ERRORS

- **Not producing sufficient vertical force from the front foot to push out of the squat position and clear the hurdle with full extension in the hip and knee**
- **Absorbing too much of the landing impact into the hip and knee of the front leg, increasing ground contact time**



31

SIDE-TO-SIDE PUSH-OFF

Intensity level: Medium

Direction: Vertical

Equipment: Plyometric box, 10 to 18 inches (25-46 cm) high

Starting position: Stand to one side of the plyometric box with one foot on the ground and one foot on the box. The inside of the foot on the box should be near the box's closest edge.

Arm action: Double arm

Preparatory movement: None

Upward movement: Jump up and over the box using the foot on the box to push off.

Downward movement: Land with the opposite foot on the opposite side of the top of the box; this foot should land just before the ground foot. Immediately repeat the movement to the opposite side.

COMMON ERRORS

- **Not producing sufficient vertical force from the front foot**
- **Poor coordination or strength deficiencies, which mean that peak height of the center of mass does not occur at the same time as maximal extension of the lower limbs and flexion in the shoulders**



32 BOX JUMP

Intensity level: Low

Direction: Vertical and slightly horizontal

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Facing the plyometric box, get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Forcefully triple extend to jump onto the top of the box using both legs.

Downward movement: Land on both feet in a half-squat position, step down from the box, and repeat.

Note: The arm action contributes significantly to the upward momentum of this jump. By having athletes perform the movement with their hands on their hips, or their fingers interlocked behind the head and elbows pointing sideways, the lower body power required to perform the jump can be significantly increased.

COMMON ERRORS

- **Focusing on landing on the box and flexing hips and knees early, thus reducing vertical power; the aim should be to land on the box because of powerful extension into the ground, maximizing the forward and upward movement of the body**
- **Not coordinating arm actions with the forward movement of the body; powerful shoulder flexion should be simultaneous with the lower body jumping action**



33

BOX HOP

Intensity level: Low

Direction: Vertical and slightly horizontal

Equipment: Plyometric box, 6 to 42 inches (15-107 cm) high

Starting position: Facing the plyometric box, get into a comfortable, upright stance on one foot. The non-jumping leg is held in a stationary position with the hip and knee flexed during the exercise.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

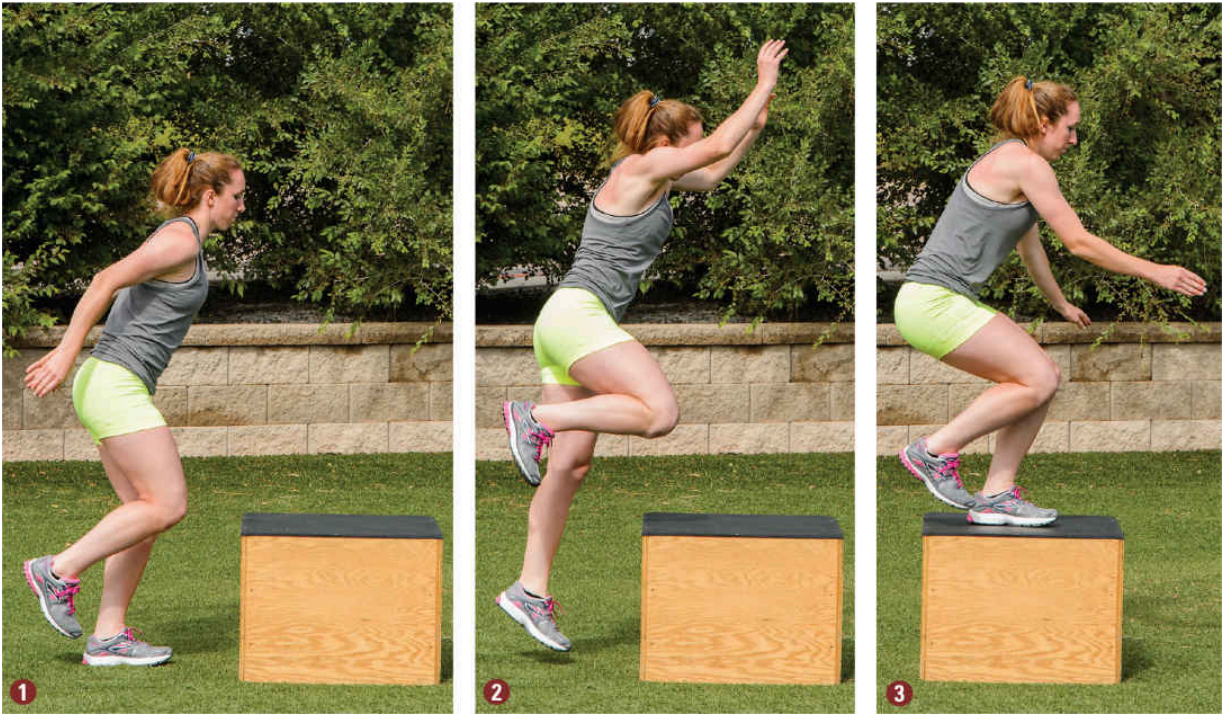
Upward movement: Forcefully triple extend and hop onto the top of the box.

Downward movement: Land on the same foot in a triple flexion position (hip, knee, and ankle flexed and stable), step down from the box, and repeat.

Note: Although significant force may be required to hop onto boxes in excess of 30 inches (76 cm), this is a low-intensity drill because the athlete is accelerating against gravity and lands with less force than his or her body weight because of this.

COMMON ERRORS

- **Focusing on landing on the box and flexing hips and knees early, thus reducing vertical power; the aim should be to land on the box because of powerful extension into the ground, maximizing the forward and upward movement of the body**
- **Not coordinating arm actions with the forward movement of the body; powerful shoulder flexion should be simultaneous with the lower body jumping action**



34 LATERAL BOX JUMP

Intensity level: Medium

Direction: Vertical and slightly horizontal

Equipment: Plyometric box, 6 to 42 inches (15-107 cm) high

Starting position: Stand to one side of the plyometric box; get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Jump onto the top of the box using both legs.

Downward movement: Land on both feet in a half-squat position and jump down, landing with an active flat-foot contact and optimal ground contact to enable the jump to be repeated in the opposite direction.

Note: If the athlete steps down rather than jumps down from the box, the drill would be classified as a low-intensity ballistic exercise. Also, this drill can be made into a low-intensity ballistic drill by stepping down off the box and jumping up and laterally only.

COMMON ERRORS

- **Focusing on landing on the box and flexing hips and knees early, thus reducing vertical power; the aim should be to land on the box because of powerful extension into the ground (and flexion of the shoulders), maximizing the forward and upward movement of the body**
- **Not leaning into the jump; bringing the center of mass outside the base of support in the frontal plane will assist the lateral action of the jump**



35 DROP FREEZE

Intensity level: Low

Direction: Vertical

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: None

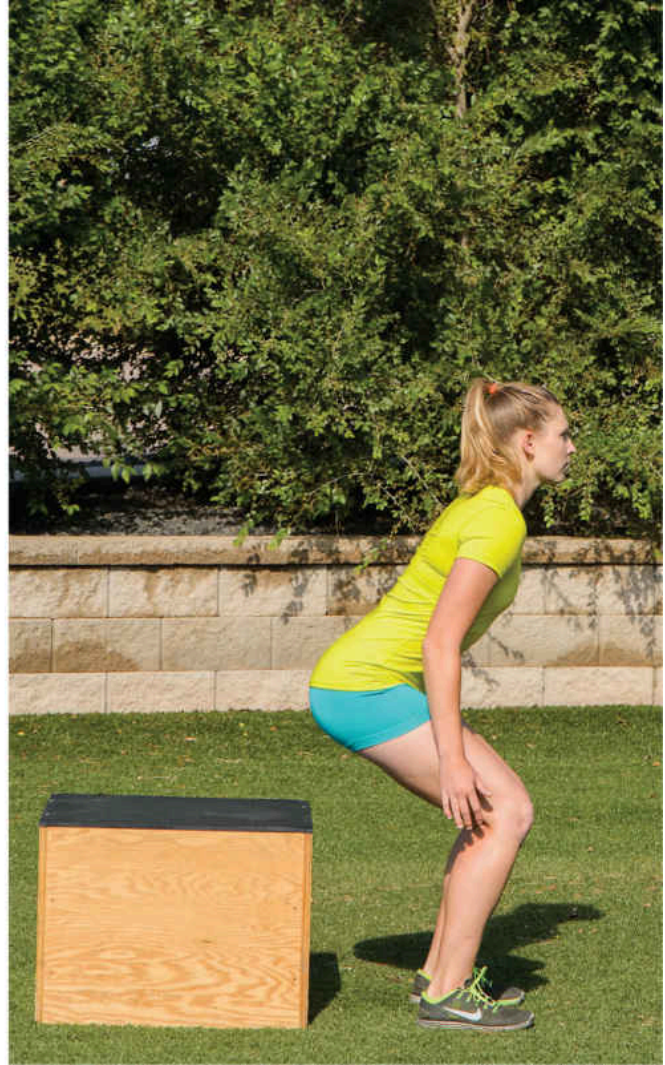
Preparatory movement: Step from the box.

Downward movement: Land flat footed on the ground with both feet, quickly absorbing the impact on landing. Ensure the hips, knees, and toes align, with the trunk vertical. Step back onto the box and repeat.

Note: This drill teaches joint alignment and landing mechanics and is often the first drill used in plyometric teaching and rehabilitation progressions. Also, this can also be performed as a single-leg landing drill.

COMMON ERRORS

- **Absorbing ground reaction forces while landing through hip and knee flexion, rather than landing and instantaneously holding position to develop load tolerance**
- **Losing the knee alignment with the toes when landing in a knee valgus position, potentially compromising the knee joint**



Depth Jumps

36 DEPTH JUMP

Intensity level: High

Direction: Vertical

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box.

Downward movement: Dorsiflex the ankles and land on the ground with an active flat-foot contact with both feet.

Upward movement: On landing, immediately jump up as high as possible.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is performed. The time on the ground should be kept to a minimum; there is an optimal relationship between ground contact time and resultant vertical force production. On landing, emphasis should be on jumping up, with minimal horizontal movement (less than what is shown in the third photo).

COMMON ERRORS

- **Absorbing ground reaction force as the athlete lands through hip and knee flexion, which increases the amortization phase and ground contact time; hips, knees, and ankles should be stiff when landing to optimize the ground contact time and rebound jump height**
- **Not dorsiflexing the ankle in preparation for ground contact; this negates the active flat-foot contact and increases ground contact time**



37 DEPTH JUMP TO BOX JUMP

Intensity level: High

Direction: Vertical and horizontal

Equipment: Two plyometric boxes, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box, facing the second box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box.

Downward movement: Dorsiflex the ankles in flight in preparation for an active flat-foot landing with both feet.

Upward movement: On landing, immediately jump onto the second box.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is performed. The time on the ground should be kept to a minimum;

there is an optimal relationship between ground contact time and resultant vertical force production. Also, the distance between boxes depends on experience and ability; the greater the distance between boxes, the higher the intensity of the jump. Begin with the boxes placed 24 inches (61 cm) apart.

COMMON ERRORS

- **Absorbing ground reaction force as the athlete lands through hip and knee flexion, which increases the amortization phase and ground contact time; hips, knees, and ankles should be stiff when landing to optimize the ground contact time and rebound jump height**
- **Not dorsiflexing the ankle in preparation for ground contact; this negates active flat-foot contact and increases ground contact time**



38 DEPTH JUMP INTO DIRECTIONAL SPRINT

Intensity level: High

Direction: Vertical and lateral

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high; a partner

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing with both feet. Have a partner point to the right or left just before landing.

Upward movement: On landing, immediately sprint in the direction determined by the partner.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is performed. The time on the ground should be kept to a minimum.

COMMON ERRORS

- **The athlete prioritizing the sprint over landing mechanics, leading to poor alignment on landing**
- **The partner signaling which direction to sprint too late for the athlete to prepare for landing and a reactionary sprint; body lean and weight distribution between the inside and outside of the feet are important considerations for successful sprints from landing**



39 DEPTH JUMP TO LONG JUMP

Intensity level: High

Direction: Vertical and horizontal

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box.

Downward movement: Dorsiflex the ankles in preparation for a flat-foot landing on both feet.